# LaTeX

PENG

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# 1 Introduction

`https://latex-project.org/`

# 2 TikZ

## 2.1 Introduction

`https://tikz.dev/`

**Definition 2.1** (TikZ)**.** *TikZ is a frontend of PGF.*

## 2.2 Curved Path Construction

**Definition 2.2** (control points)**.** `<ps> .. controls <pc1> and <pc2> .. <pe>`

```
\begin{tikzpicture}[scale=2]
    \filldraw [gray]
    (0,0) circle (2pt)
    (1,1) circle (2pt)
    (2,1) circle (2pt)
    (2,0) circle (2pt);
    \draw [thick] (0,0) .. controls (1,1) and (2,1) .. (2,0);
    \draw [-latex] (0,0) -- node[sloped,above] {tangent} (1,1);
    \draw [-latex] (2,0) -- node[sloped,below] {tangent} (2,1);
\end{tikzpicture}
```

